MILNOR SECTION 2 SUPPLEMENT

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1. Hessian of a Function on Manifolds

Let M be a smooth n-manifold and $f \in C^\infty(M, \mathbb{R})$. Recall that one can define the differential:

$$df_p \colon T_pM \longrightarrow \mathbb{R}.$$

In local coordinates $(x^1, \dots, x^n)$, it is represented by a vector

$$df_p = \left(\frac{\partial f}{\partial x^1}(p), \dots, \frac{\partial f}{\partial x^m}(p)\right)$$

A point $p \in M$ is *critical* when $df_p = 0$. The second derivative is more subtle. In local coordinates $(x^1, \dots, x^m)$, one may attempt to define its *Hessian* at p naive by

$$\mathrm{Hess}_p(f) = \left(\frac{\partial^2 f}{\partial x^i \partial x^j}(p)\right)_{1\le i,j\le m}. \tag{1.1}$$

However, if if $x = x(y)$ is a change of coordinates, then by chain rule:

$$\frac{\partial^2 f}{\partial y^a \partial y^b}(p) = \frac{\partial x^i}{\partial y^a}(p)\frac{\partial x^j}{\partial y^b}(p)\frac{\partial^2 f}{\partial x^i \partial x^j}(p) + \frac{\partial^2 x^i}{\partial y^a \partial y^b}(p)\frac{\partial f}{\partial x^i}(p). \tag{1.2}$$

Therefore, $\mathrm{Hess}_p(f)$ is not a tensor, i.e., depending on the choice of local coordinates. As we have seen in Milnor's text, the annoying extra term vanishes when p is a critical point of f:

Proposition 1.3. *Suppose $df_p = 0$. For $v, w \in T_pM$, extending to vector fields V, W near p and define*

$$\mathrm{Hess}_p\, f(v, w) := V(Wf)(p).$$

This is independent of choice of V, W and is a symmetric bilinear form on T_pM. In coordinates, it is given by

$$\left(\frac{\partial^2 f}{\partial x^i \partial x^j}(p)\right)_{i,j}.$$

Remark 1.4. Milnor [1] calls $[V, W]$ a "Poisson bracket"; in current terminology this is the *Lie bracket* of vector fields. The Hessian is nondegenerate if the induced map $T_pM \to T_p^*M$ is an isomorphism. Its index is the maximal dimension of a subspace on which it is negative definite. Sylvester's law of inertia shows that the index is coordinate-independent.

Question 1.1. *How to define Hessian of f as a bilinear form on tangent spaces for non-critical points?*

Recall that one definition of tangent spaces T_pM of M at p is given by

$$T_pM = \{\gamma : (-\varepsilon, \varepsilon) \to M : \gamma(0) = p\}/\sim,$$

where $\gamma_1 \sim \gamma_2$ if $\dot\gamma_1(0) = \dot\gamma_2(0)$. In this formulation, if $\gamma(0) = p$ and $\dot\gamma(0) = v$, then we can compute:

$$\frac{d^2}{dt^2}\Big|_{t=0} f(\gamma(t)) = \frac{\partial^2 f}{\partial x^i \partial x^j}\dot\gamma^i(0)\dot\gamma^j(0) + \frac{\partial f}{\partial x^i}\ddot\gamma^i(0).$$

Hence, we see that the difficulties of defining Hessian are.

(1) The acceleration term $\ddot{\gamma}(0)$ depends on the choice of coordinates (same calculation as (1.2)).
(2) Curves with the same initial velocity may have different accelerations. Indeed, on $M = \mathbb{R}$, consider $\gamma_1(t) = t$ and $\gamma_2(t) = t + t^2$. Then $\dot{\gamma}_1(0) = \dot{\gamma}_2(0) = 0$ but $\ddot{\gamma}_1(0) = 0 \neq 2 = \ddot{\gamma}_2(0)$.

When $df_p = 0$, the troubling acceleration vanishes:

$$\mathrm{Hess}_p f(v,v) = \frac{d^2}{dt^2}\bigg|_{t=0} f(\gamma(t)) =$$

For general case, we need to introduce an *affine connection* ∇ on TM. With this, we define

$$(\mathrm{Hess}^\nabla f)(X,Y) := (\nabla df)(X,Y) = X(Yf) - (\nabla_X Y)f. \tag{1.5}$$

It is tensorial in X, Y, and in coordinates

$$(\mathrm{Hess}^\nabla f)_{ij} = \frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma^k_{ij}\frac{\partial f}{\partial x^k}.$$

The Christoffel-symbol term exactly cancels the extra term in (1.2). If ∇ is torsion-free, then $\mathrm{Hess}^\nabla f$ is symmetric. If two connections differ by $\widehat{\nabla}_X Y = \nabla_X Y + A(X,Y)$, then

$$\mathrm{Hess}^{\widehat{\nabla}} f(X,Y) - \mathrm{Hess}^\nabla f(X,Y) = -df(A(X,Y)).$$

Consequently all connections give Milnor's intrinsic Hessian at a critical point.

If g is Riemannian, its Levi–Civita connection gives the *Riemannian Hessian*

$$\mathrm{Hess}_g f := \nabla^g df, \qquad \mathrm{Hess}_g f(X,Y) = g(\nabla^g_X \operatorname{grad}_g f, Y).$$

For the geodesic γ_v with $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v$,

$$\mathrm{Hess}_g f|_p(v,v) = \frac{d^2}{dt^2}\bigg|_0 f(\gamma_v(t)). \tag{1.6}$$

In normal coordinates at p, the Christoffel symbols vanish at p, so the Riemannian Hessian is represented there by the ordinary matrix of second partials, even when p is not critical.

2. Hadamard Lemma and Zariski Tangent Spaces

We first recall Hadamard lemma as in [1, Lemma 2.1]:

Lemma 2.1. *Let V be a convex open neighborhood of* 0 *in $\mathbb{R}^m$ and $f \in C^\infty(V)$ with $f(0) = 0$. Then*

$$f(x_1,\ldots,x_m) = \sum_{j=1}^m x_j g(x_1,\ldots,x_m) \tag{2.2}$$

for some $g_i \in C^\infty(V)$ with $g_i(0) = \frac{\partial f}{\partial x_i}(0)$.

Proof. The proof is a straightforward application of fundamental theorem of calculus and chain rule:

$$f(x_1,\ldots,x_m) = \int_0^1 \frac{d}{dt} f(tx_1,\ldots,tx_m)dt = \int_0^1 \sum_{j=1}^m \frac{\partial f}{\partial x_i}(tx_1,\ldots,tx_m)x_i dt.$$

We then take $g_i(x) = \int_0^1 \frac{\partial f}{\partial x_i}(tx_1,\ldots,tx_m)dt$ and $g_i(0) = \frac{\partial f}{\partial x_i}(0)$ as expected. □

The elementary Lemma 2.1 also has an important application to tangent spaces.

Recall that the tangent space of a smooth manifold M at p is defined by equivalence classes of curves: consider smooth curves $\gamma : (-\varepsilon, \varepsilon) \to M$ satisfying $\gamma(0) = p$. We declare

$$\gamma_1 \sim \gamma_2 \iff \frac{d}{dt}(\phi \circ \gamma_1)(0) = \frac{d}{dt}(\phi \circ \gamma_2)(0) \quad \text{in some local chart } \phi \text{ around } p,.$$

On the other hand, if $f \in C^\infty(U)$ and $g \in C^\infty(V)$ are defined on some neighborhoods U, V of p, then we define an equivalence relation:

$$f \sim g \iff f|_W = g|_W \text{ for some } W \subset U \cap V.$$

We denote the resulting equivalence class by f_p, called the *germ of $f \in C^\infty(U)$ at p*. We denote $C^\infty_{M,p}$ the local algebra of germs of smooth functions at p.

Definition 2.3. A *derivation* of $C^\infty_{M,p}$ is a $\mathbb{R}$-linear map $D : C^\infty_{M,p} \to \mathbb{R}$ satisfying the Leibniz rule

$$D(fg) = f(p)D(g) + g(p)D(f).$$

The vector space of such derivations is denoted by $\mathrm{Der}_{\mathbb{R}}(C^\infty_{M,p}, \mathbb{R})$.

A curve γ through p defines a derivation by

$$D_\gamma(f) := \frac{d}{dt}\bigg|_{t=0} f(\gamma(t)).$$

We now use Lemma 2.1 to show that every derivation arises in this way.

Lemma 2.4. $\mathrm{Der}_{\mathbb{R}}(C^\infty_{M,p}, \mathbb{R}) \cong T_pM$.

Proof. Choose a local coordinates $x^1, \ldots, x^n$ near p, so that $x^i(p) = 0$. By Lemma 2.1, every smooth germ f can be written as

$$f - f(p) = \sum_{i=1}^n x^i g_i, \qquad g_i(p) = \frac{\partial f}{\partial x^i}(p).$$

Applying a derivation D and using the Leibniz rule gives

$$D(f) = \sum_{i=1}^n D(x^i) g_i(p) = \sum_{i=1}^n D(x^i) \frac{\partial f}{\partial x^i}(p).$$

Thus D is directional differentiation by the tangent vector

$$v_D := \sum_{i=1}^n D(x^i)\ \frac{\partial}{\partial x^i}\bigg|_p.$$

Consequently, $\mathrm{Der}_{\mathbb{R}}(C^\infty_{M,p}, \mathbb{R}) \simeq T_pM$. □

There is also an algebraic formulation. Let

$$\mathfrak{m}_p = \{f \in C^\infty_{M,p} : f(p) = 0\}$$

be the maximal ideal. Every derivation annihilates constants and $\mathfrak{m}_p^2$, so

$$\mathrm{Der}_{\mathbb{R}}(C^\infty_{M,p}, \mathbb{R}) \simeq \left(\mathfrak{m}_p/\mathfrak{m}_p^2\right)^*.$$

Lemma 2.1 shows that the map

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \longrightarrow T_p^*M, \qquad [f] \longmapsto df_p,$$

is an isomorphism. Hence, we see that the tangent space coincides with *Zariski tangent space*.

$$\boxed{T_pM \simeq \mathrm{Der}_{\mathbb{R}}(C^\infty_{M,p}, \mathbb{R}) \simeq \left(\mathfrak{m}_p/\mathfrak{m}_p^2\right)^*.}$$

The preceding algebraic description is special to C^∞. For a C^k-function, Lemma 2.1 still gives

$$f(x) - f(0) = \sum_{i=1}^{n} x^i g_i(x), \qquad g_i(x) = \int_0^1 \frac{\partial f}{\partial x^i}(tx)\,dt,$$

but in general g_i is only C^{k-1}. Thus factoring out a coordinate may lose one derivative, and the factorization does not remain inside the ring of C^k-functions. Moreover, one can show that:

Question 2.1. *If $k \in \mathbb{N}^*$, Zariski tangent space $\mathfrak{m}_p/\mathfrak{m}_p^2$ is infinite-dimensional!*

The usual tangent space of a C^k-manifold is still perfectly well-defined using curves or coordinate velocities. On the other hand, Zariski tangent space is usually the way to define tangent spaces for singular spaces (e.g., in algebraic geometry).

3. Various Forms of Morse Lemma

Lemma 3.1 (Morse Lemma)**.** *Let p be a non-degenerate critical point for f. Then there is a local coordinate system $(y^1, \dots, y^n)$ in a neighborhood U of p with $y^i(p) = 0$ for all i and such that the identity*

$$f = f(p) - (y^1)^2 \cdots - (y^\lambda)^2 + (y^{\lambda+1})^2 + \cdots + (y^n)^2.$$

holds throughout U, where λ is the index of f at p.

In [1, Lemma 2.2], we prove this by induction on n. We can also prove this simultaneously.

Second Proof of Morse inequality. Taylor's formula with integral remainder gives

$$f(x) = f(0) + \frac{1}{2}\langle B(x)x, x\rangle \qquad B(x) = 2\int_0^1 (1-s)D^2 f(sx)ds \tag{3.2}$$

Thus $B(x)$ is symmetric and $B(0) = H$. We wish to find $\psi(t) = R(t)t$, where $R(0) = I$, satisfying

$$R(t)^* B(R(t)t)R(t) = H. \tag{3.3}$$

We define a map $\mathcal{F}(t, R) : \mathbb{R}^n \times M_n(\mathbb{R}) \to M_n(\mathbb{R})$ by

$$\mathcal{F}(t, R) = R^* B(Rt)R - H$$

with values in the vector space $\mathrm{Sym}_n(\mathbb{R})$ of symmetric matrices. At $(0, I)$,

$$D_R\mathcal{F}(0, I)(K) = K^* H + HK.$$

This map $M_n(\mathbb{R}) \to \mathrm{Sym}_n$ is surjective. Indeed, for $S \in \mathrm{Sym}_n(\mathbb{R})$, i.e., $S = S^*$,

$$K = \frac{1}{2}H^{-1}S \quad \Longrightarrow \quad K^* H + HK = S.$$

The existence of the function $R(t)$ is now a consequence of implicit function theorem. We take

$$R = I + \frac{1}{2}H^{-1}S, \qquad S \in \mathrm{Sym}_n.$$

On this slice, the derivative of $\mathcal{F}$ in S at the origin is the identity. The implicit function theorem gives a smooth $S(t)$, hence $R(t)$, solving (3.3). Since $R(0) = I$, the map $\psi(t) = R(t)t$ has $D\psi(0) = I$ and is locally invertible. Substitution into (3.2) proves that

$$f(\psi(t)) = f(0) + \frac{1}{2}\langle Ht, t\rangle.$$

Finally, by Sylvester's law, a constant linear change transforms $\frac{1}{2}H$ to $\mathrm{diag}(-I_\lambda, I_{n-\lambda})$. □

Remark 3.4. If $f \in C^k$, $k > 2$, B in (3.2) is C^{k-2}, and this proof produces a C^{k-2} coordinate change.

This second proof is more abstract but it's more useful when one want to deal with *Morse function with parameter*. Suppose $F(s,x)$ is a smooth family of functions, where $s \in \mathbb{R}^m$ is a parameter.

Theorem 3.5. *Let $F\colon P \times U \to \mathbb{R}$ be smooth near $(s_0, x_0) \in \mathbb{R}^m \times \mathbb{R}^n$. Assume*

$$D_x F(s_0, x_0) = 0, \qquad D_x^2 F(s_0, x_0) \text{ is nonsingular of index } \lambda.$$

After shrinking the neighborhoods, there are:

(i) a unique smooth critical section $c(s)$, with $c(s_0) = x_0$ and $D_x F(s, c(s)) = 0$;
(ii) a parameter-preserving local diffeomorphism $(s,y) \mapsto (s, \Phi(s,y))$, with $\Phi(s,0) = c(s)$, such that

$$F(s, \Phi(s,z)) = F(s, c(s)) - \sum_{j=1}^{\lambda} (y^j)^2 + \sum_{j=\lambda+1}^{n} (y^j)^2. \tag{3.6}$$

Proof. Apply the implicit function theorem to the map $(s,x) \mapsto D_x F(s,x)$. Its derivative in x at (s_0, x_0) is the invertible vertical Hessian, so it produces the unique smooth critical section $c(s)$.

Set $G(s,u) = F(s, c(s) + u)$. Then $D_u G(s,0) = 0$. Repeating (3.2) gives

$$G(s,u) = G(s,0) + \frac{1}{2}\langle B(s,u)u, u\rangle.$$

In the implicit function theorem argument, s may be retained as an additional parameter. We obtain a smooth family of coordinate changes reducing the phase to its vertical Hessian $H(s) = D_x^2 F(s, c(s))$.

These Hessians remain nonsingular nearby. Apply the same sliced matrix argument to

$$(s, A) \longmapsto A^* H(s) A - H(s_0)$$

to obtain a smooth congruence with the fixed $H(s_0)$. A final constant linear change gives (3.6). □

The parametric form of Morse lemma shows the more general *Morse–Bott lemma*.

Corollary 3.7. *Let $N \subset M$ be an embedded submanifold consisting of critical points of f, and suppose*

$$\ker \operatorname{Hess}_p f = T_p N \qquad (p \in N).$$

Near every $p_0 \in N$, there are coordinates (u, y^-, y^+) in which

$$N = \{y^- = y^+ = 0\}, \qquad f = f(p_0) - |y^-|^2 + |y^+|^2.$$

The dimensions of the positive and negative normal spaces are locally constant on C.

Proof. Choose submanifold coordinates (u,y) with $C = \{y = 0\}$. The condition on the kernel says precisely that the Hessian in the normal y-variables is nondegenerate. Regard u as a parameter and apply Theorem 3.5. Finally, $f|_N$ is locally constant because $df = 0$ along C. □

Corollary 3.8. *A nondegenerate critical point survives a sufficiently small smooth perturbation, uniquely and with the same index.*

4. Application of Morse Lemma

Following Hörmander [3, §VII.7], we now give an application of Morse Lemma to analysis.

Let $a \in C_c^\infty(\mathbb{R}^n)$, $\phi \in C^\infty(\mathbb{R}^n; \mathbb{R})$, we consider the *oscillatory integral*:

$$I(\omega) = \int_{\mathbb{R}^n} e^{i\omega\phi(x)} a(x) dx.$$

where a is called the *ampltitude* and ϕ is called the *phase*. Our goal is to understand the asymptotic behavior of the above integral as $\omega \to \infty$. The first observation is that:

Lemma 4.1 (Non-Stationary phase lemma)**.** *If f $d\phi \neq 0$ on* $\operatorname{supp} a$*, then*

$$I(\omega) = O(\omega^{-N}) \qquad \textit{for every } N.$$

Thus regions without critical points make no asymptotic contribution.

Proof. On a neighborhood of $\operatorname{supp} a$, define

$$V = \frac{D\phi}{|D\phi|^2}.$$

Since $\langle V, D\phi\rangle = 1$, we have $\langle V, D(e^{i\omega\phi})\rangle = i\omega e^{i\omega\phi}$. Integration by parts therefore gives

$$I(\omega) = -\frac{1}{i\omega}\int_{\mathbb{R}^n} e^{i\omega\phi} \operatorname{div}(aV)\, dx.$$

Set $a_{j+1} = \operatorname{div}(a_j V)$. Each $a_j \in C_c^\infty(\mathbb{R}^n)$ and supported in $\operatorname{supp} a$. Repeating the argument yields

$$I(\omega) = \frac{(-1)^N}{(i\omega)^N}\int_{\mathbb{R}^n} e^{i\omega\phi} a_N\, dx,$$

and hence $|I(\omega)| \leq \omega^{-N}\|a_N\|_{L^1}$. □

Thus regions without critical points contribute only a rapidly decaying error. To find the asymptotic term, it remains to study the integral near critical points of ϕ.

Theorem 4.2 (Stationary phase formula)**.** *Assume that x_0 is the only critical point of ϕ on* $\operatorname{supp} a$*, and*

$$H = D^2\phi(x_0)$$

is nonsingular. Let λ be its index and $\sigma = n - 2\lambda$ its signature. Then there exist coefficients $c_j \in \mathbb{C}$, depending only on finitely many derivatives of a and ϕ at x_0, such that, for every integer $N \geq 1$,

$$I(\omega) = \left(\frac{2\pi}{\omega}\right)^{n/2} e^{i\omega\phi(x_0)} e^{i\pi\sigma/4}\left(\sum_{j=0}^{N-1} c_j\omega^{-j} + O(\omega^{-N})\right), \tag{4.3}$$

as $\omega \to +\infty$, where $c_0 = \frac{a(x_0)}{\sqrt{|\det H|}}$.

Proof of the leading term. We prove only the case $N = 1$. The Morse lemma reduces the phase to a quadratic form, and the Gaussian integral determines its contribution.

Choose a cutoff equal to 1 near x_0 and supported in a Morse coordinate neighborhood. By Lemma 4.1, the remaining part of the integral is $O(\omega^{-N})$ for every N. The Morse lemma, in its Hessian form, gives a local change of coordinates $x = \psi(y)$ satisfying

$$\psi(0) = x_0, \qquad D\psi(0) = I, \qquad \phi(\psi(y)) = \phi(x_0) + \frac{1}{2}\langle Hy, y\rangle.$$

Absorbing the cutoff and the absolute Jacobian determinant into the amplitude, we need to estimate

$$J(\omega) = \int_{\mathbb{R}^n} e^{i\omega\langle Hy,y\rangle/2} b(y)\, dy,$$

where $b \in C_c^\infty(\mathbb{R}^n)$ and $b(0) = a(x_0)$.

We first compute the corresponding Gaussian integral. The familiar formula

$$\int_{\mathbb{R}} e^{-zt^2/2 + i\eta t}\, dt = \left(\frac{2\pi}{z}\right)^{1/2} e^{-\eta^2/(2z)}, \qquad \operatorname{Re} z > 0,$$

follows from the real Gaussian integral and analytic continuation; the square root is chosen positive when $z > 0$. For $h \in \mathbb{R} \setminus \{0\}$, setting $z = \varepsilon - i\omega h$ and letting $\varepsilon \downarrow 0$ gives

$$\lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}} e^{-\varepsilon t^2/2 + i\omega h t^2/2 + i\eta t}\, dt = \left(\frac{2\pi}{\omega |h|}\right)^{1/2} e^{i\pi \operatorname{sgn}(h)/4} e^{-i\eta^2/(2\omega h)}.$$

Diagonalizing H and multiplying these one-dimensional formulas therefore yields

$$\lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} e^{-\varepsilon |y|^2/2 + i\omega \langle Hy, y\rangle/2 + iy\cdot\xi}\, dy = \left(\frac{2\pi}{\omega}\right)^{n/2} \frac{e^{i\pi\sigma/4}}{\sqrt{|\det H|}} e^{-i\langle H^{-1}\xi, \xi\rangle/(2\omega)}.$$

To handle the amplitude, write

$$\widehat{b}(\xi) = \int_{\mathbb{R}^n} e^{-iy\cdot\xi} b(y)\, dy, \qquad b(y) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{iy\cdot\xi} \widehat{b}(\xi)\, d\xi.$$

Insert this expression into $J(\omega)$, first with the Gaussian factor $e^{-\varepsilon|y|^2/2}$. Fubini's theorem and the preceding calculation, followed by dominated convergence as $\varepsilon \downarrow 0$, give

$$J(\omega) = \left(\frac{2\pi}{\omega}\right)^{n/2} \frac{e^{i\pi\sigma/4}}{\sqrt{|\det H|}} \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{-i\langle H^{-1}\xi, \xi\rangle/(2\omega)} \widehat{b}(\xi)\, d\xi.$$

Here dominated convergence is justified because the regularized Gaussian integrals are bounded uniformly in ξ and $\varepsilon > 0$, for fixed $\omega > 0$, and $\widehat{b}$ is rapidly decreasing.

Finally,

$$\left| e^{-i\langle H^{-1}\xi, \xi\rangle/(2\omega)} - 1 \right| \le \frac{C_H |\xi|^2}{\omega}.$$

Since $|\xi|^2 \widehat{b}(\xi)$ is integrable, Fourier inversion at 0 gives

$$\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{-i\langle H^{-1}\xi, \xi\rangle/(2\omega)} \widehat{b}(\xi)\, d\xi = b(0) + O(\omega^{-1}).$$

Restoring the factor $e^{i\omega\phi(x_0)}$ and using $b(0) = a(x_0)$, we obtain

$$I(\omega) = \left(\frac{2\pi}{\omega}\right)^{n/2} e^{i\omega\phi(x_0)} e^{i\pi\sigma/4} \frac{a(x_0)}{\sqrt{|\det H|}} + O(\omega^{-n/2-1}),$$

as claimed. ☐

Remark 4.4. If $\phi(x, s)$ depends smoothly on a parameter and has a nondegenerate critical point $x = c(s)$, the parameterized Morse lemma chooses the quadratic coordinates smoothly in s. Therefore the same stationary-phase formula holds uniformly for s in a small compact set, with x_0, H, and $a(x_0)$ replaced by $c(s)$, $D^2_{xx}\phi(c(s), s)$, and $a(c(s), s)$.

The Gaussian calculation above also gives two classical formulas.

Corollary 4.5 (Fresnel integrals)**.** *We have*

$$\int_0^\infty \cos(t^2)\, dt = \int_0^\infty \sin(t^2)\, dt = \sqrt{\frac{\pi}{8}}.$$

Proof. For $\varepsilon > 0$, the Gaussian formula gives

$$\int_0^\infty e^{-(\varepsilon - i)t^2}\, dt = \frac{\sqrt{\pi}}{2} (\varepsilon - i)^{-1/2}.$$

Letting $\varepsilon \downarrow 0$, we obtain

$$\int_0^\infty e^{it^2}\,dt = \frac{\sqrt{\pi}}{2}e^{i\pi/4}.$$

The limit agrees with the ordinary improper integral: integration by parts gives a tail bound $O(R^{-1})$ on $[R,\infty)$, uniformly for $\varepsilon \geq 0$. Taking real and imaginary parts proves the result. $\square$

For Stirling's formula, the relevant analogue of stationary phase is *Laplace's method*. Instead of cancellation away from a critical point, exponential decay localizes the integral near a minimum.

Suppose that f has a nondegenerate minimum at t_0, with $f(t_0) = 0$ and $f''(t_0) > 0$, and that b is smooth and compactly supported sufficiently near t_0. Then

$$\int e^{-\omega f(t)} b(t)\,dt = \sqrt{\frac{2\pi}{\omega f''(t_0)}}\big(b(t_0) + O(\omega^{-1})\big).$$

Indeed, the Morse lemma gives a coordinate y such that

$$f(t(y)) = \frac{y^2}{2}, \qquad t'(0) = \frac{1}{\sqrt{f''(t_0)}}.$$

The transformed amplitude has a Taylor expansion

$$B(y) = B(0) + B'(0)y + O(y^2), \qquad B(0) = \frac{b(t_0)}{\sqrt{f''(t_0)}}.$$

After localizing with an even cutoff, the odd term integrates to zero, while

$$\int_{\mathbb{R}} e^{-\omega y^2/2}\,dy = \sqrt{\frac{2\pi}{\omega}}, \qquad \int_{\mathbb{R}} y^2 e^{-\omega y^2/2}\,dy = O(\omega^{-3/2}).$$

The part away from $y = 0$ is exponentially small, proving the formula.

Corollary 4.6 (Stirling's formula)**.** *As $n \to \infty$,*

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \big(1 + O(n^{-1})\big).$$

Proof. Repeated integration by parts gives

$$n! = \int_0^\infty e^{-x} x^n\,dx.$$

Substituting $x = nt$, we obtain

$$n! = n^{n+1}e^{-n}\int_0^\infty e^{-nf(t)}\,dt, \qquad f(t) = t - 1 - \log t.$$

The function f has a unique minimum at $t = 1$, where

$$f(1) = f'(1) = 0, \qquad f''(1) = 1.$$

Thus Laplace's method gives

$$\int_0^\infty e^{-nf(t)}\,dt = \sqrt{\frac{2\pi}{n}}\big(1 + O(n^{-1})\big).$$

To justify localization despite the noncompact interval, note that f is bounded below by a positive constant on $(0,2]$ outside any fixed neighborhood of 1. Moreover, for $t \geq 2$,

$$f'(t) \geq \frac{1}{2}, \qquad \int_2^\infty e^{-nf(t)}\,dt \leq \frac{2}{n}e^{-nf(2)}.$$

Hence all contributions away from the minimum are exponentially small. Multiplying by $n^{n+1}e^{-n}$ proves the formula. □

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